A source-centred ridge for heteroscedastic probit models with few target rows

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Abstract

A heteroscedastic probit, a probit whose noise scale depends on features, cannot pin down its scale coefficients from a few hundred target rows. We pull its ridge penalty towards a fit on a large related source sample, and we pick the strength by grouped cross-validation on a grid that holds both endpoints: target-only and source-only. On a public football era-shift task this beats the plain target-only fit by -0.0265 nats of log-loss per row (SE 0.0077) at 50 target games and by -0.0056 (SE 0.0014) with the full 284-game season; on a men's-to-women's basketball task it ties. But it does not beat source-only, which is better by +0.0058 (SE 0.0025) on football at the full season and by +0.0324 (SE 0.0145) on basketball at 25 games, so our pre-registered test (beat target-only, source-only and pooled at once) failed on both real datasets. We pair every comparison on later held-out seasons, average it over 20 target subsamples per size and bootstrap whole test games.

1 Problem

Many binary outcomes are threshold crossings of a quantity that keeps moving. A game ends with the home team ahead or not; a level ends above a mark or not. The classic answer is a Brownian closed form [Stern, 1994]: with signed distance d to the threshold, remaining horizon h and base scale s, the probability is $\Phi(d/(s\sqrt{h}))$. Its weak spot is the scale. How far the quantity can still move depends on the state of the game, so a heteroscedastic probit [Harvey, 1976] learns the scale as a log-linear function of features, as weather post-processing does [Gneiting et al., 2005, Messner et al., 2014].

We hit the limit of this model when the target sample is small. With a few hundred target rows the scale coefficients are not identified, and a plain ridge shrinks them towards zero, which is towards the closed form again. Often a large related sample exists: earlier seasons, another league, another product. Fitting on it alone ignores how the target differs. Pooling both needs the source rows at fit time. We ask whether a ridge pulled towards the source fit, instead of towards zero, beats all three.

2 The change

We keep the model fixed and change only the penalty. For row i the fitted scale, the score and the probability are

$$\sigma_i = s_i \exp(a + w^\top z_i), \qquad \eta_i = \frac{u_i}{\sigma_i} + b, \qquad P(y_i = 1) = F\big(\mathrm{clip}(\eta_i, -c, c)\big), \qquad u_i = \frac{d_i}{\sqrt{h_i}}. \tag{1}$$

With parameters $\theta = (a, w, b)$ set to zero, this is the closed form of Section 1. The plain variant penalises towards zero, and we penalise towards the source fit θ_0:

$$\hat{\theta}_{\text{plain}}(\lambda) = \arg\min_\theta \ \ell_T(\theta) + \tfrac{\lambda}{2}\big(\|w\|^2 + b^2\big), \qquad \hat{\theta}(\lambda) = \arg\min_\theta \ \ell_T(\theta) + \tfrac{\lambda}{2}\|\theta - \theta_0\|^2. \tag{2}$$

Preprint.

Symbol	Meaning
d_i, h_i, s_i	signed distance to the threshold, remaining horizon, base scale
z_i	log scale features, standardised with the *source* mean and sd
a, w, b	scale intercept, scale slopes, bias of the score
F, c	standard normal CDF (or Student-t), clip bound $c = 12$
ℓ_T	summed negative log-likelihood on the target rows
θ_0	the source fit (plain variant with $\lambda = 1$ on the source)

Three choices make it more than a prior mean. First, the penalty covers every coordinate, including the scale intercept a, which the plain variant leaves free, because the overall level of the scale is the term a small target estimates worst. Second, we freeze the standardisation of z at the source values, so θ_0 means the same thing on both samples. Third, λ comes from grouped cross-validation (folds that keep all rows of a game together) over $\{0, 1, 10, 100, 1000, 10000, \infty\}$ with 5 folds, and ties go to the larger λ. Both endpoints are on the grid: $\lambda = \infty$ returns θ_0 exactly (source-only), and $\lambda = 0$ is the unpenalised target fit started at θ_0 (target-only). The CV table therefore shows directly whether adapting to the target paid. We fit with L-BFGS-B [Byrd et al., 1995], a quasi-Newton solver with box bounds, using an analytic gradient and boxes $|a| \le 1.5$, $|w_j|, |b| \le 2$. The gradient is zero on clipped rows, where the loss is flat; a non-zero gradient there stopped the optimiser after one step in an early build.

None of this is new in general. A ridge centred at a source fit is MAP adaptation (a prior centred at the old model, Chelba and Acero, 2006), biased regularisation [Kuzborskij and Orabona, 2013] and L2-SP [Li et al., 2018], and Tian and Feng [2023] study GLM transfer with a CV-chosen strength. Our delta is narrow: the heteroscedastic probit with the three choices above, measured against both endpoints. Proposition 1 restates the classic local argument for this likelihood; the proof is in Appendix A.

Proposition 1 (local risk, after Kuzborskij and Orabona, 2013)**.** *Take the quadratic approximation of ℓ_T around the target optimum θ_*, with total Fisher information (the curvature of the log-likelihood) $J \succ 0$ from n target rows, penalty $\frac{\lambda}{2}(\theta - \theta_0)^\top \Lambda (\theta - \theta_0)$ with Λ diagonal, $\Lambda \succeq 0$, $\Lambda \neq 0$, and gap $\delta = \theta_0 - \theta_*$. The expected excess risk $R(\lambda)$ in the per-row metric J/n has slope $R'(0) = -2\,\mathrm{tr}(J^{-1}\Lambda)/n < 0$. So some $\lambda > 0$ beats target-only for any finite δ, and for $\Lambda \succ 0$, $R(\lambda) \to \delta^\top (J/n)\,\delta$ as $\lambda \to \infty$.*

So a small pull always helps in the quadratic regime. The proposition does not say cross-validation finds the right λ on a few dozen groups, and the likelihood is not convex, so we claim no unique optimum.

3 Setup

Data. Football (play-by-play from Ho and Carl, 2024): source seasons 2009–2017 (2,398 games, 47,960 rows), target pool season 2021 (284 games), test seasons 2022–2024 (852 games, 17,040 rows). The distance is the home margin plus the pregame point spread times the share of the game left (Stern's drift); the 4 scale features are the remaining time, the margin, timeouts left and the pregame total-points line. Basketball [Gilani and Hutchinson, 2024]: source men's league 2019–2024 (7,565 games, 151,300 rows), target pool women's league 2019–2022 (817 games), test 2023–2025 (838 games, 16,760 rows), 3 scale features (time, pace, margin). Each game gives 20 random game states, and $y = 1$ when the home team wins. The test seasons come after every training season, so the football task is an era shift (train on old seasons, test on new ones), and nothing tunes on them. A synthetic generator with known truth gives a regime map over 20 cells (shift $\times$ target rows, 20 draws each).

Arms. Closed form ($\theta = 0$); target-only (plain variant, λ by the same CV); source-only; pooled (source and target rows with a domain indicator in the scale); ours. Generic learners: logistic regression [Pedregosa et al., 2011] target-only and pooled, LightGBM (gradient-boosted trees, Ke et al., 2017) target-only, source-only and fine-tuned from the source booster. 5 pre-registered ablations: λ fixed at 1, 100 and 10000, a Student-$t(4)$ link, and target standardisation. Every arm sees the same target rows; logistic regression picks its penalty and LightGBM its boosting rounds by the same grouped CV.

Metric and SE. The primary metric is held-out log-loss $L = -\frac{1}{N}\sum_i [y_i \log p_i + (1-y_i)\log(1-p_i)]$ in nats per row. Target sizes are 25, 50, 100, 200 (and 400 for basketball) games, 20 random subsamples each, plus the full pool. The paired SE of $L_{\text{ours}} - L_{\text{arm}}$ is a block bootstrap over test games (2,000 draws), resampling whole games because rows of one game are dependent, plus the between-subsample variance divided by 20. We call a difference inside 1.96 SE a tie. Before the run we pre-registered the primary test: ours against target-only, source-only and pooled at every subsampled size, with success if all three upper 95% bounds are below zero on either real dataset.

4 Results

Ours beats target-only on football and does not beat source-only anywhere. Table 1 gives the full-season fit, where cross-validation picked $\lambda = 1000$ for ours on both datasets, a finite value, so the fit did move away from the source.

Table 1: Ours beats the target-only fit on football and loses to source-only and pooling there; on basketball every scaleprobit arm ties. Full target season, held-out log-loss (nats per row, lower is better) and the paired difference ours − arm with its SE; negative means ours is better.

	Football (284 target games)		Basketball (817 target games)	
Arm	log-loss	ours − arm (SE)	log-loss	ours − arm (SE)
Closed form	0.4908	−0.0202 (0.0054)	0.4732	−0.0016 (0.0028)
Target-only	0.4762	−0.0056 (0.0014)	0.4719	−0.0004 (0.0006)
Source-only	0.4648	+0.0058 (0.0025)	0.4709	+0.0006 (0.0022)
Pooled	0.4652	+0.0054 (0.0021)	0.4716	−0.0000 (0.0007)
Ours	**0.4706**		**0.4715**	
Logistic reg., target	0.4751	−0.0045 (0.0016)	0.4715	+0.0000 (0.0004)
Logistic reg., pooled	0.4678	+0.0028 (0.0021)	0.4711	+0.0005 (0.0018)
LightGBM, target	0.5094	−0.0388 (0.0076)	0.4782	−0.0067 (0.0012)
LightGBM, source	0.4628	+0.0078 (0.0046)	0.4699	+0.0016 (0.0023)
LightGBM, fine-tuned	0.4760	−0.0055 (0.0055)	0.4705	+0.0010 (0.0014)

On football ours lowers the target-only log-loss from 0.4762 to 0.4706, a paired gain of −0.0056 (SE 0.0014). But source-only scores 0.4648 and pooling 0.4652, both better than ours beyond their SE. Our reading is that the finite λ lets the fit chase quirks of the single target season, and those quirks are gone in the three test seasons that follow it. LightGBM fitted on the source alone has the lowest log-loss on both datasets (0.4628 and 0.4699); ours ties it. Basketball is different. The women's league is close to Brownian: the closed form alone scores 0.4732 against 0.4715 for ours, so the scale has little left to learn.

Figure 1 shows the paired difference against the three pre-registered rivals at every target size; Table 2 in Appendix B gives the numbers.

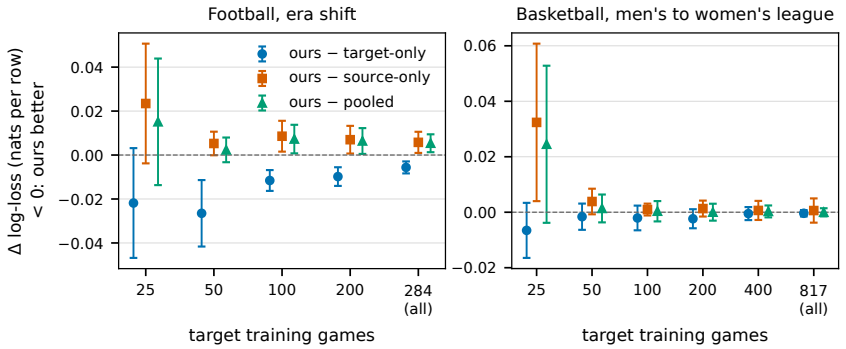


Figure 1: Ours beats target-only on football from 50 games on, but source-only and pooling sit at or below it from 50 games on; on basketball everything ties except the 25-game loss to source-only. Paired held-out log-loss difference ours − rival, bars are 95% intervals (±1.96 SE).

The largest football gain over target-only is -0.0265 (SE 0.0077) at 50 games; at 25 games the mean gain -0.0218 is inside its SE of 0.0127, a tie. Against the uncorrected source fit the sign never flips: from 100 games on that endpoint wins beyond SE, and at 50 games it sits right at the edge (+0.0053, SE 0.0027). The worst basketball case is the smallest target, where the centred ridge trails source-only by +0.0324 (SE 0.0145), because CV on 25 games picked $\lambda = \infty$ in only 8 of 20 subsamples and $\lambda = 0$ in 1. So the pre-registered success criterion holds at 0 sizes on football and 0 on basketball.

Calibration tells the same story (Figure 2). On football the expected calibration error (ECE, Guo et al., 2017, the mean gap between forecast and observed win rates over 15 equal-width bins) is 0.0326 for target-only, 0.0297 for ours and 0.0155 for source-only.

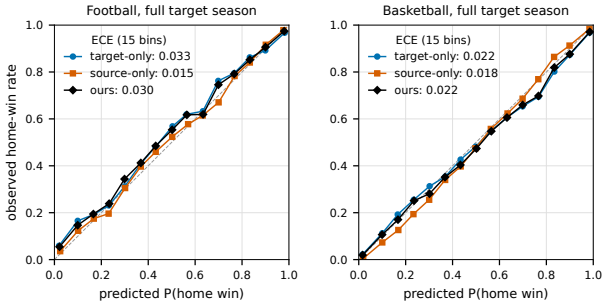


Figure 2: Source-only is the best calibrated arm on both datasets; ours sits between it and target-only. Reliability diagrams at the full target season, bins with at least 30 rows, ECE in the legend.

Synthetic data, where the true shift is known, shows when adaptation pays (Figure 3). The centred ridge improves on target-only beyond SE in 15 of 20 cells and never loses to it. It overtakes source-only in 5 cells, all at shift 0.3 or 0.6 with enough target rows, and falls behind in 4, where the shift is small. All three rivals fall at once in 1 cell (shift 0.3, 800 rows: -0.0012, -0.0091, -0.0077). Which endpoint wins depends on the shift. With a small target, grouped CV cannot tell a real shift from noise, and its choice between the endpoints is close to a coin flip.

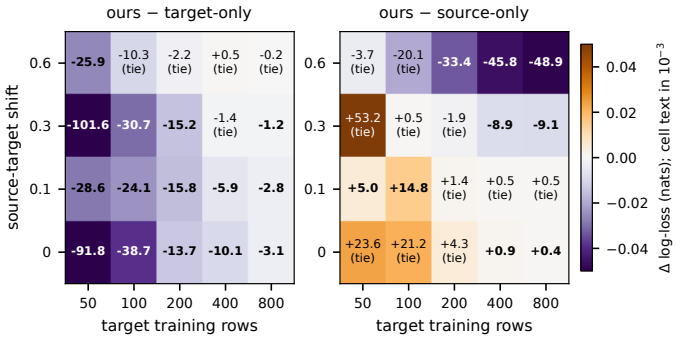


Figure 3: Ours beats target-only almost everywhere but beats source-only only when the shift is large and the target has enough rows. Synthetic regime map, paired log-loss difference over 20 draws; plain text marks a tie.

5 Limits

The pre-registered success test failed on both real datasets. The method reliably improves on the plain target fit, but on these tasks the best single choice was to use the source fit as it is. We do not claim the method beats source-only or pooling, nor that it generalises beyond these two sports and the synthetic generator.

After the main run we tried 4 post-hoc variants of the λ selection: a 13-point grid, slopes pulled far harder than the intercept, the one-standard-error rule and repeated CV (Appendix B). None was kept. The one-SE rule picked $\lambda = \infty$ in every real subsample, so it is source-only by another name, and it loses on synthetic cells where the shift is real. The 5 ablations were pre-registered. Appendix C lists the cuts and where the method came from.

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A Proof of Proposition 1

Given. 1) Quadratic approximation: $\ell_T(\theta) \approx \ell_T(\hat\theta) + \frac{1}{2}(\theta-\hat\theta)^\top J(\theta-\hat\theta)$ with $J \succ 0$ and $\hat\theta$ the unpenalised target fit. 2) $J(\hat\theta - \theta_*) = g$ with $\mathbb{E}[g] = 0$ and $\mathrm{Cov}(g) = J$. 3) Objective $Q_\lambda(\theta) = \ell_T(\theta) + \frac{\lambda}{2}(\theta-\theta_0)^\top \Lambda(\theta-\theta_0)$, Λ diagonal, $\Lambda_{jj} \geq 0$, at least one $\Lambda_{jj} > 0$ (the package uses $\Lambda = I$). 4) Risk $R(\lambda) = \mathbb{E}\big[(\hat\theta(\lambda)-\theta_*)^\top (J/n)(\hat\theta(\lambda)-\theta_*)\big]$. 5) Matrix rules: $\frac{d}{d\lambda}A^{-1} = -A^{-1}\frac{dA}{d\lambda}A^{-1}$; $\mathrm{tr}(XY) = \mathrm{tr}(YX)$; $\mathbb{E}[g^\top M g] = \mathrm{tr}(MJ)$ for $\mathbb{E}[g]=0$, $\mathrm{Cov}(g) = J$.

Choose: $A = J + \lambda\Lambda$ and $m = \lambda A^{-1}\Lambda\delta$.

Need to prove: $\hat\theta(\lambda) - \theta_* = m + A^{-1}g$.

$$
\begin{aligned}
0 &= J(\theta-\hat\theta) + \lambda\Lambda(\theta-\theta_0) && \text{gradient of } Q_\lambda, \text{ given 1 and 3}\\
A\theta &= J\hat\theta + \lambda\Lambda\theta_0 && \text{collect the terms in } \theta\\
A(\theta-\theta_*) &= J(\hat\theta-\theta_*) + \lambda\Lambda(\theta_0-\theta_*) && \text{subtract } A\theta_* = J\theta_* + \lambda\Lambda\theta_*\\
\theta-\theta_* &= A^{-1}g + \lambda A^{-1}\Lambda\delta && \text{given 2, multiply by } A^{-1}
\end{aligned}
$$

The last line needs A invertible, therefore, for $x \neq 0$:

$$
\begin{aligned}
x^\top A x &= x^\top J x + \lambda\, x^\top \Lambda x && \text{definition of } A\\
&> 0 && J \succ 0,\ \lambda \geq 0,\ \Lambda \succeq 0
\end{aligned}
$$

Same method, on the risk. Substitution in given 4:

$$
\begin{aligned}
R(\lambda) &= \mathbb{E}\big[(m + A^{-1}g)^\top \tfrac{J}{n}(m + A^{-1}g)\big] && \text{first part}\\
&= m^\top \tfrac{J}{n} m + 2\, m^\top \tfrac{J}{n} A^{-1}\mathbb{E}[g] + \mathbb{E}\big[g^\top A^{-1}\tfrac{J}{n}A^{-1}g\big] && \text{expand, } m \text{ is not random}\\
&= \underbrace{\lambda^2 \delta^\top \Lambda A^{-1}\tfrac{J}{n}A^{-1}\Lambda\delta}_{B(\lambda)} + \underbrace{\tfrac{1}{n}\mathrm{tr}(A^{-1}JA^{-1}J)}_{V(\lambda)} && \mathbb{E}[g]=0, \text{ given 5}
\end{aligned}
$$

Need to prove: $R'(0) = -2\,\mathrm{tr}(J^{-1}\Lambda)/n$. $B(\lambda) = \lambda^2\beta(\lambda)$ with β smooth at 0, so $B'(0) = 2\cdot 0\cdot\beta(0) + 0 = 0$. For the variance:

$$
\begin{aligned}
V'(\lambda) &= \tfrac{1}{n}\big[\mathrm{tr}(-A^{-1}\Lambda A^{-1}JA^{-1}J) + \mathrm{tr}(A^{-1}J(-A^{-1}\Lambda A^{-1})J)\big] && \text{product rule, given 5}\\
V'(0) &= -\tfrac{1}{n}\big[\mathrm{tr}(J^{-1}\Lambda J^{-1}JJ^{-1}J) + \mathrm{tr}(J^{-1}JJ^{-1}\Lambda J^{-1}J)\big] && A = J \text{ at } \lambda = 0\\
&= -\tfrac{1}{n}\big[\mathrm{tr}(J^{-1}\Lambda) + \mathrm{tr}(J^{-1}\Lambda J^{-1}J)\big] && J^{-1}J = I\\
&= -\tfrac{2}{n}\mathrm{tr}(J^{-1}\Lambda) && \text{cyclic trace, given 5}
\end{aligned}
$$

The sign needs $\mathrm{tr}(J^{-1}\Lambda) > 0$, therefore:

$$
\begin{aligned}
\mathrm{tr}(J^{-1}\Lambda) &= \textstyle\sum_j (J^{-1})_{jj}\Lambda_{jj} && \Lambda \text{ diagonal}\\
&= \textstyle\sum_j (e_j^\top J^{-1} e_j)\,\Lambda_{jj} && \text{unit vectors } e_j\\
&> 0 && J^{-1} \succ 0, \text{ all } \Lambda_{jj} \geq 0, \text{ one} > 0
\end{aligned}
$$

Therefore $R'(0) = B'(0) + V'(0) = -2\,\mathrm{tr}(J^{-1}\Lambda)/n < 0$, and some $\lambda > 0$ has $R(\lambda) < R(0)$, the target-only risk.

Same method, on the limit ($\Lambda \succ 0$).

$$\begin{aligned} \lambda A^{-1} = (J/\lambda + \Lambda)^{-1} &\to \Lambda^{-1} && \text{divide } A \text{ by } \lambda \\ m = \lambda A^{-1}\Lambda\delta &\to \delta, \qquad A^{-1} \to 0 && \text{first line} \\ R(\lambda) &\to \delta^\top \tfrac{J}{n}\delta + 0 && \text{substitution in } B \text{ and } V \end{aligned}$$

That is the source-only risk. □

B Learning curve, ablations and post-hoc variants

Table 2 gives the numbers behind Figure 1.

Table 2: No size meets the pre-registered test: ours never beats source-only, and only beats target-only on football. Paired held-out log-loss difference ours − rival (SE) per target size; **bold** is a win beyond 1.96 SE.

target games	ours − target-only	ours − source-only	ours − pooled
Football			
25	−0.0218 (0.0127)	+0.0234 (0.0139)	+0.0151 (0.0147)
50	**−0.0265 (0.0077)**	+0.0053 (0.0027)	+0.0023 (0.0029)
100	**−0.0116 (0.0024)**	+0.0086 (0.0036)	+0.0073 (0.0033)
200	**−0.0098 (0.0022)**	+0.0070 (0.0032)	+0.0064 (0.0030)
all 284	**−0.0056 (0.0014)**	+0.0058 (0.0025)	+0.0054 (0.0021)
Basketball			
25	−0.0065 (0.0051)	+0.0324 (0.0145)	+0.0245 (0.0144)
50	−0.0016 (0.0024)	+0.0039 (0.0024)	+0.0014 (0.0026)
100	−0.0021 (0.0023)	+0.0010 (0.0011)	+0.0004 (0.0019)
200	−0.0023 (0.0018)	+0.0013 (0.0015)	+0.0000 (0.0016)
400	−0.0005 (0.0012)	+0.0007 (0.0018)	+0.0003 (0.0011)
all 817	−0.0004 (0.0006)	+0.0006 (0.0022)	−0.0000 (0.0007)

Table 3 removes one design choice at a time at the full season, and Table 4 compares the 4 post-hoc selection variants with the default on the same target subsamples.

Table 3: Cross-validated λ beats fixed small λ on football but loses to the fixed large one, which sits next to source-only; source standardisation helps on football. Full target season, ours − ablation (SE).

Ablation	Football	Basketball
$\lambda = 1$ fixed	−0.0110 (0.0025)	−0.0011 (0.0006)
$\lambda = 100$ fixed	−0.0084 (0.0020)	−0.0009 (0.0005)
$\lambda = 10000$ fixed	+0.0051 (0.0019)	+0.0013 (0.0014)
Student-$t(4)$ link	+0.0012 (0.0011)	−0.0001 (0.0005)
Target standardisation	−0.0014 (0.0006)	+0.0003 (0.0002)

Table 4: No post-hoc variant beats the default by enough to keep; the one-SE rule wins at a few small sizes only because it always returns source-only. Variant − default (SE) on the same subsamples.

Variant	25 games	50 games	100 games
Football			
13-point grid	+0.0003 (0.0093)	+0.0011 (0.0018)	−0.0019 (0.0018)
Slopes anchored	+0.0214 (0.0170)	−0.0012 (0.0030)	−0.0004 (0.0030)
One-SE rule	−0.0234 (0.0139)	−0.0053 (0.0027)	−0.0086 (0.0036)
Repeated CV	−0.0183 (0.0134)	+0.0071 (0.0073)	−0.0043 (0.0024)
Basketball			
13-point grid	+0.0057 (0.0054)	−0.0004 (0.0015)	+0.0007 (0.0006)
Slopes anchored	−0.0078 (0.0068)	+0.0010 (0.0033)	+0.0013 (0.0011)
One-SE rule	−0.0324 (0.0145)	−0.0039 (0.0024)	−0.0010 (0.0011)
Repeated CV	−0.0120 (0.0094)	+0.0007 (0.0024)	−0.0002 (0.0008)

C Origin, reproduction and cuts

Origin. We first built the method for a private application, and this is the one statement we make about it. On a proprietary binary forecasting task (a few hundred target training examples; a related source sample more than an order of magnitude larger), centring the ridge at the source fit improved held-out log-loss by 0.0056 nats (0.3161 → 0.3105) and ECE from 0.030 to 0.027 over the target-only fit chosen by an extensive cross-validated search (paired block-bootstrap SE of the log-loss delta 0.0032, so about 1.7 SE; the 95 % interval, -0.0120 to +0.0007, includes 0). Cross-validation chose a lambda so large that the fit stayed within 0.003 of the source parameters: at this target size the gain came from the source fit, and target adaptation added nothing measurable. Against an earlier, lightly tuned target-only configuration the difference was within noise (paired SE 0.0005-0.0009).

Reproduction. The package is `scaleprobit` (Python, SciPy L-BFGS-B, Virtanen et al., 2020). The benchmark runs on CPU in 856 s (football), 1,376 s (basketball) and 2,628 s (synthetic). Commands, from the repository root: `python -m bench.run_football`, `bench.run_hoops`, `bench.run_synthetic`, `bench.variants.run football|hoops`, `bench.variants.synthetic_one_se`; then `paper/results/refit_full.py`, `paper/results/make_numbers.py` and `paper/figures/make.py` regenerate every number and figure in this note. The full-season refit reproduces the benchmark log-loss of every scaleprobit arm, and its per-game SE matches the block SE at the full season (0.0014 against 0.0014 on football). Cuts, none silent: a third real dataset; the R arms `glmx::hetglm` [Zeileis et al., 2026] and `glmtrans`; the Platt, free-intercept and grouped-λ ablations; the football source starts later than planned to bound download and fit time. Basketball target files carry no usable pregame spread, so neither side uses Stern's drift there. LightGBM uses one fixed parameter set; only its rounds are chosen by grouped CV.